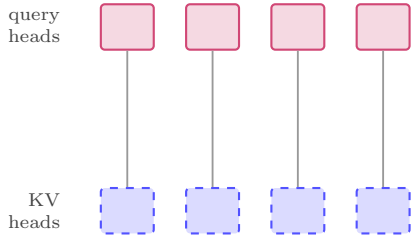


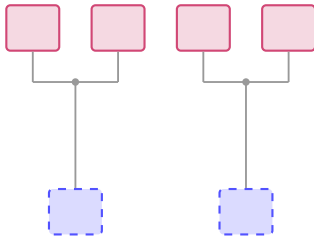
Sharing key/value heads

MHA



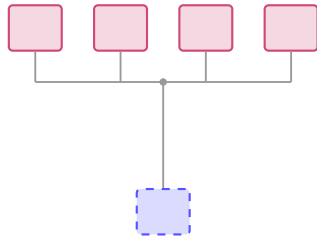
$$\#KV = h = 4$$

GQA



$$\#KV = g = 2$$

MQA



$$\#KV = 1$$

$$KV \text{ cache} \propto \#KV \text{ heads: } 2n h d_k \text{ (MHA)} \cdot 2n g d_k \text{ (GQA)} \cdot 2n d_k \text{ (MQA)}$$

queries stay per-head (only $\mathbf{K}$, $\mathbf{V}$ are shared); fewer distinct KV heads $\Rightarrow$ fewer cache reads per generated token — the decode-bandwidth win
 (example $h=4$, $g=2 \Rightarrow 1 : \frac{1}{2} : \frac{1}{4}$ of the MHA cache)